Practical No. 08

Title of the Assignment: Develop a Simple Chatbot Using Predefined Rules and Pattern Matching

Problem statement: Develop a simple chatbot that interacts with a user using predefined rules and pattern matching. The chatbot should identify keywords or patterns in the user's input and generate an appropriate predefined response.

Objective:

- To understand the basic concept of chatbots in Artificial Intelligence.
- To understand rule-based conversation systems.
- To implement pattern matching for identifying user inputs.
- To develop a simple chatbot using predefined rules and responses.

Theory:

Chatbot:

A chatbot is a computer program that interacts with users through text or voice. A simple rule-based chatbot does not learn from data. Instead, it uses predefined rules to identify patterns in the user's input and returns a corresponding predefined response.

Rule-Based Chatbot:

In a rule-based chatbot, the conversation is controlled using a collection of IF–THEN rules. The input message is first processed to identify words or patterns. The chatbot compares these patterns with predefined rules. If a matching rule is found, its associated response is displayed. If no rule matches, a default response is generated.

Pattern Matching:

Pattern matching is the process of checking whether the user's input contains a particular word, phrase or pattern. For example, if the input contains the word 'hello', the chatbot can match it with a greeting rule and respond with 'Hello! How can I help you?'.

Basic Working of the Chatbot:

1. User enters a message.
2. The chatbot converts the input into a suitable form, such as lowercase text.
3. The chatbot checks the input against predefined patterns.
4. If a pattern matches, the corresponding response is selected.
5. If multiple patterns are available, the first applicable rule can be selected.
6. If no pattern matches, the chatbot gives a default response.
7. The conversation continues until the user enters an exit command such as 'bye'.

Input Data / Predefined Rule Data:

The following rules can be used for implementing the chatbot:

Rule	Input Pattern	Predefined Response
Rule 1	If input contains: hello, hi, hey	Hello! How can I help you?
Rule 2	If input contains: name	I am a simple AI chatbot.
Rule 3	If input contains: how are you	I am fine. Thank you for asking!
Rule 4	If input contains: course, subject	I can provide basic information about your course.
Rule 5	If input contains: help	Sure! You can ask me about my name, course, or general information.
Rule 6	If input contains: thank, thanks	You are welcome!
Rule 7	If input contains: bye, exit, quit	Goodbye! Have a nice day.
Rule 8	If no pattern matches	Sorry, I do not understand your question.

Example Conversation:

User: Hello

Chatbot: Hello! How can I help you?

User: What is your name?

Chatbot: I am a simple AI chatbot.

User: Can you help me?

Chatbot: Sure! You can ask me about my name, course, or general information.

User: Thank you

Chatbot: You are welcome!

User: Bye

Chatbot: Goodbye! Have a nice day.

Algorithm:

8. Step 1: Start.
9. Step 2: Define the predefined patterns and corresponding responses.
10. Step 3: Display a welcome message.
11. Step 4: Accept the user's input.
12. Step 5: Convert the input into lowercase to make matching case-insensitive.
13. Step 6: Check whether the input contains an exit pattern such as 'bye', 'exit' or 'quit'.
14. Step 7: If an exit pattern is found, display the exit response and stop.
15. Step 8: Otherwise, compare the input with each predefined pattern.
16. Step 9: If a pattern matches, display the corresponding predefined response.

17. Step 10: If no pattern matches, display the default response.
18. Step 11: Repeat Steps 4–10 for the next user input.
19. Step 12: Stop.

Pseudo Code:

```
START
Define rules and responses
Display welcome message
REPEAT
   Read user_input
   Convert user_input to lowercase
   IF user_input matches exit pattern:
      Display goodbye response
      STOP
   ELSE:
      matched = FALSE
      FOR each rule:
         IF input matches rule pattern:
            Display corresponding response
            matched = TRUE
            BREAK
      IF matched == FALSE:
         Display default response
UNTIL chatbot is stopped
END
```

Sample Input and Output:

User Input	Chatbot Output
Hi	Hello! How can I help you?
What is your name?	I am a simple AI chatbot.
Can you help me?	Sure! You can ask me about my name, course, or general information.
What is the subject?	I can provide basic information about your course.
Good morning	Sorry, I do not understand your question.
Bye	Goodbye! Have a nice day.

Advantages and Limitations of Rule-Based Chatbots:

Advantages:

- Simple and easy to understand.
- Easy to implement using IF–THEN rules.
- Does not require a large training dataset.
- Responses are predictable because they are predefined.
- Useful for simple FAQ and basic customer-support applications.

Limitations:

- Cannot understand complex natural language effectively.
- Cannot generate new responses outside the predefined rules.
- Performance depends on the number and quality of predefined patterns.
- Different ways of asking the same question may not match the same rule.
- It does not learn automatically from previous conversations.

Conclusion:

A simple rule-based chatbot was developed using predefined rules and pattern matching. The chatbot accepts user input, identifies matching keywords or patterns, and provides a predefined response. This practical demonstrates the basic working of conversational AI and provides an introduction to more advanced chatbot techniques such as Natural Language Processing and machine-learning-based dialogue systems.

Oral Questions:

1. What is a chatbot?
20. What is a rule-based chatbot?
21. What is pattern matching?
22. How are rules represented in a simple chatbot?
23. Why is the input converted to lowercase?
24. What happens when no predefined pattern matches the user's input?
25. What is the purpose of a default response?
26. What is the difference between a rule-based chatbot and an AI/ML-based chatbot?
27. What are the advantages of a rule-based chatbot?
28. What are the limitations of a rule-based chatbot?